# COURSE WORK NAZIH

01 / COMPLETE RESEARCH ROADMAP - 29 OCTOBER 2026

## OBJECTIVE

Industrial Motor: one-image decision, Normal (0) or Defective (1).
Dataset: public M2AD Motor. Course: Deep Learning for Computer Vision.

## PHASE A - ORIGINAL EXPLORATION

Started with M2AD Motor as a metal quality / anomaly-detection case.
Checked the Motor images, masks, illumination pairing, and test boundaries.
Initial Ring examples were only a technical smoke test, not Motor evidence.

## PHASE B - ANOMALY METHODS

Ran frozen ResNet18 feature-based reference and a documented adaptation
of the core AnomalyDINO computational idea.
Used normal Motor reference samples; checked whether scores and code worked.
This was anomaly scoring, not a trained two-class supervised classifier.

## PHASE C - TWO ILLUMINATIONS

Tested two views of the same part under different lighting.
Compared mean-score fusion, max-score fusion and joined features.
Added a learned residual between the illumination representations.
Checked it on 6 normal parts; raised the maximum score for 2 of 6.
This could increase false alarms: not yet evidence of improvement.

# WHAT THE OLD EXPERIMENTS SHOWED

02 / ACTUAL RESULTS AND WHAT THEY MEAN

## HISTORICAL FINAL ANOMALY EVALUATION

Max fusion: Average Precision (AP) = 94.12%.
Mean fusion: AP = 95.20%.
Max plus residual: AP = 95.23%.
Compared with Max: +1.11 percentage points, confidence interval -0.64 to +3.22 points, so statistically unproven superiority.
The localization score was slightly worse than for Max fusion.

## IMPORTANT SCIENTIFIC LIMITS

AP is not classifier accuracy; a ranked anomaly score is not by itself a Normal / Defective class decision.
Old Motor evaluation images were exposed to research decisions.
We cannot call a new split of those same images a blind final test.

## PROFESSOR’S REQUIREMENTS RECHECKED

The project should classify one image, with at least TWO classes.
It requires a baseline, one central paper idea, same-data comparison, metrics, failure cases, reproducible code and final B-BRICS report.
The anomaly experiments remain supplementary, not discarded.

# THE CORRECTED SUPERVISED PROJECT



## PHASE D - VERIFIED SUPERVISED BASELINE

Selected single-image Motor A000 / I01 with labels 0 Normal and 1 Defective.
Trained all ResNet18 parameters using Cross-Entropy on labelled images.
Pilot: 99 images total; 70 train and 29 validation.
Training classes: 35 Normal / 35 Defective.
Validation classes: 15 Normal / 14 Defective.
Validation accuracy: 28/29 = 96.55%; macro-F1 = 96.54%.
These are development results, NOT independent test metrics.

## PHASE E - CONTROLLED PAPER EXPERIMENT

Selected PolyLoss (Leng et al., ICLR 2022), Poly-1 correction:
Loss = CrossEntropy + epsilon * (1 - probability_of_true_class).
Set epsilon = 1.0; keep same ResNet18, initial weights, split,
optimizer and training budget. The only changed factor is the loss.
Code is prepared. Poly-1 training has NOT been completed.

## NEXT STEPS

1. Open 03_ALL_CODE.ipynb in a Colab GPU runtime.
2. Fetch the original 99 images and verify every SHA256.
3. Run Cross-Entropy and Poly-1 matched training; record actual metrics.
4. Audit physical specimen independence and get a proper holdout.
5. Later write the final report and slides, not now.

## SCHEDULE

30 Oct plan / 6 Nov baseline / 13 Nov paper comparison.
20 Nov metrics / 27 Nov report draft / 4 Dec final defense.